VITAMIN C

Scurvy

↑ iron absorption

CHEMICAL NAME (5R)-5-[(1S)-1,2-dihydroxyethyl]-3,4-dihydroxyfuran-2(5H)-one (L-ascorbic acid)
CLASS Water-soluble vitamin (antioxidant)
TRADE NAMES Redoxon · Cebion · Celin · Ascorbic acid injection

THERAPEUTIC USES / INDICATIONS

- Scurvy (treatment and prevention).
- Vitamin C deficiency: malnutrition, alcohol misuse, smokers, elderly with poor diet, dialysis.
- Enhance absorption of oral iron.
- Methaemoglobinaemia (when methylene blue unavailable/contraindicated); wound healing support in deficiency.

ADMINISTRATION

- **PO** tablets, effervescent or chewable, with food.
- **IV/IM** in severe deficiency or malabsorption (IV slowly).

DOSAGES

- **Scurvy:** at least 250 mg daily in divided doses (max 1000 mg/day adults; 500 mg/day age 4–11), then maintenance.
- **Prevention:** meet the RDA (women 75 mg, men 90 mg daily; smokers +35 mg).
- **With iron:** may be taken with iron to improve absorption (or with orange juice).

PHARMACOKINETICS

- Absorbed in small intestine (saturable >1 g).
- Concentrated in leucocytes, adrenals; metabolised to oxalate.
- Excess excreted in urine. **t½:** ~10–20 days (body pool).

PHARMACODYNAMICS

- Cofactor for **collagen synthesis** (prolyl and lysyl hydroxylases); antioxidant; reduces Fe^{3+} to Fe^{2+} → ↑ iron absorption; supports immune function.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- GI upset, diarrhoea (high doses).
- Heartburn.
- Flushing.

ADVERSE EFFECTS (SERIOUS)

- Oxalate kidney stones (high doses, renal impairment).
- Haemolysis in G6PD deficiency (high IV doses).
- Iron overload worsening in haemochromatosis.
- Interference with glucose meter readings (high IV doses).

CONTRAINDICATIONS & CAUTIONS

- Hyperoxaluria, oxalate stones.
- **Caution:** haemochromatosis, G6PD deficiency, renal impairment.

KEY INTERACTIONS

- **Iron:** ↑ absorption (beneficial; harmful in overload).
- **Aluminium antacids:** ↑ aluminium absorption.
- **Warfarin:** high doses may ↓ effect (rare).

NURSING MANAGEMENT

- Assess for scurvy signs: bleeding gums, perifollicular haemorrhages, corkscrew hairs, poor wound healing, bruising.
- Dietary assessment and education.
- Encourage fluids with high doses.

PATIENT EDUCATION

- Eat fresh fruit and vegetables daily (citrus, peppers, tomatoes).
- Do not take very high doses long-term (kidney stones).
- Take with iron tablets to help absorption.

KEY POINTS

- Vitamin C — collagen synthesis, antioxidant.
- Enhances iron absorption.
- Scurvy: 250–500 mg daily.
- High doses → oxalate stones.

SOURCES CHECKED **[943]** Ascorbic Acid 100 mg Tablets (UK SmPC) **[944]** NIH ODS vitamin C fact sheet **[945]** Ascor, IV (US label) **[946]** Ascorbic Acid, IV (US label)
Full references p. 432–457